Chapter summary

Familiarising yourself with **items in green** is important for effective revision.
Paying attention to items in the boxes is helpful for understanding the concept(s).

6.1 The Short Run and the Long Run

Variable factors and fixed factors

Variable factors	Fixed factors
They are the factors of production that **change in quantity as output changes**.	They are the factors of production that **do not change in quantity as output changes**.

Meaning of the short run and the long run

The short run	The long run
It refers to a period in which at least one of the factors of production is a fixed factor.	It refers to a period in which all factors of production are variable factors.

The short run and the long run are not defined in terms of the length of time.

6.2 Input-output Relationship in the Short Run

Total product, average product and marginal product

- **Total product (TP)** is the total quantity of output produced by a firm in a period of time.
- **Average product (AP)** is the average quantity of output produced by one unit of a variable factor in a period of time.

$$\text{Average product (AP)} = \frac{\text{Total product (TP)}}{\text{Units of a variable factor}}$$

- **Marginal product (MP)** is **the change in total product resulting from employing an additional unit of a variable factor in a period of time**.

$$\text{MP of the } n\text{th unit of a variable factor} = \text{TP of } n \text{ units of the variable factor} - \text{TP of } (n-1) \text{ units of the variable factor}$$

$$\text{Marginal product (MP)} = \frac{\text{Change in total product}}{\text{Change in a variable factor}}$$

When marginal product decreases, as long as it is positive, total product still increases but at a decreasing rate.

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Law of diminishing marginal returns

- The **law of diminishing marginal returns** states that **in the short run, as more and more units of a variable factor are added to a given quantity of fixed factor successively, marginal product will decrease eventually, other things being constant.**
- The law of diminishing marginal returns is **not applicable in the long run**.
- All other things involved in the production process are assumed to be constant and the variable factor must be homogeneous.

6.3 Cost of Production

Variable costs and fixed costs

Variable costs	They are the **costs that change as output changes**.	In both the short run and the long run, there are variable costs.
Fixed costs	They are the **costs that do not change as output changes**.	• In the short run, fixed costs are still larger than zero when the firm produces nothing. • In the long run, there are no fixed costs

Total cost, average cost and marginal cost

- **Total cost (TC)** is the sum of all costs incurred in production.

$$\text{Total cost (TC)} = \text{Total fixed cost (TFC)} + \text{Total variable cost (TVC)}$$

When the firm produces nothing in the short run, the total cost equals the total fixed cost, which is greater than zero.

- Average cost (AC) and average variable cost (AVC)
  - **Average cost (AC)** is the total cost per unit of output.

$$\text{Average cost (AC)} = \frac{\text{Total cost (TC)}}{\text{Output (Q)}}$$

  - **Average variable cost (AVC)** is the total variable cost per unit of output.

$$\text{Average variable cost (AVC)} = \frac{\text{Total variable cost (TVC)}}{\text{Output (Q)}}$$

- **Marginal cost (MC)** is **the change in total cost resulting from producing an additional unit of output**.

$$\text{MC of the } n\text{th unit of output} = \text{TC of producing } n \text{ units of output} - \text{TC of producing } (n-1) \text{ units of output}$$

$$\text{Marginal cost (MC)} = \frac{\text{Change in total cost } (\Delta TC)}{\text{Change in output } (\Delta Q)}$$

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6.4 Economies and Diseconomies of Scale



When a firm enjoys economies of scale in production, it is producing in the long run.

- When a firm increases (reduces) its output level by increasing (reducing) the employment of all factors of production, its scale of production increases (decreases).
- **Economies of scale:** Average cost decreases when the scale of production increases.
- **Diseconomies of scale:** Average cost increases when the scale of production further increases.
- **Optimal scale:** The firm produces at an output level where its average cost is at minimum in the long run.
- The long run average cost (LRAC) curve is U-shaped.

Internal economies and diseconomies of scale



When internal economies of scale occur in a firm, the firm's profit does not necessarily increase.

- **Internal economies of scale** occur in a firm when **an increase in production scale causes a decrease in long run average cost**.
  Possible causes are:
  - lower buying costs
  - lower cost of borrowing
  - more resources for research and development
  - usage of more advanced machinery
  - higher degree of or more extensive specialisation
  - wider spread of marketing expenses
  - lower cost in managing risks
- **Internal diseconomies of scale** occur in a firm when **an increase in production scale causes an increase in long run average cost**.
  Possible causes are:
  - communication problems
  - difficulties in management
  - huge administrative cost
  - higher marketing expenses
  - higher cost of borrowing
  - higher cost of purchase

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External economies and diseconomies of scale

- **External economies of scale** occur in an industry when **the expansion of the industry causes a decrease in long run average cost of a firm**.
  Possible causes are:
  - higher accessibility
  - availability of educated labour
  - lower sourcing costs
  - lower marketing expenses
- **External diseconomies of scale** occur in an industry when **the expansion of the industry causes an increase in long run average cost of a firm**.
  Possible causes are:
  - higher labour costs
  - higher rental costs
  - more environmental problems
  - saturation of existing market

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